2 Data and Simulated samples

2.1 Data

This study uses certified events from the Run2016BCDEFGH dataset, corresponding to 36.33 fb^{-1}, Run 2017BCDEF dataset corresponding to 41.48 fb^{-1} and Run2018ABCD dataset corresponding to 59.83 fb^{-1} [4]. The Ultralegacy datasets are used:

- 2016 Data corresponds to
  - Cert_271036-284044_13TeV_Legacy2016_Collisions16_JSON.txt
- 2017 Data corresponds to
  - Cert_294927-306462_13TeV_UL2017_Collisions17_GoldenJSON.txt
- 2018 Data corresponds to
  - Cert_314472-325175_13TeV_Legacy2018_Collisions18_JSON.txt

2.2 Trigger Strategy

Events to be used for this analysis are captured primarily by large-radius jet triggers, with a mixture of HT-based triggers included to help maximize efficiency. All triggers are unprescaled for the duration of their availability, though some triggers were not available for all run periods. An inclusive large-radius jet trigger is unprescaled only at $p_{\mathrm{T}} \gtrsim 500\,\mathrm{GeV}$, we use also some triggers that use minimum mass and/or b-tagging requirements to help lower the turn-on threshold.

We use a logical "OR" of the following triggers for 2016 data:

- `AK8DiPFJet280_200_TrimMass30_BTagCSV_p20`
- `AK8PFHT700_TrimR0p1PT0p03Mass50`
- `AK8PFJet360_TrimMass30`
- `PFHT650_WideJetMJJ900DEtaJJ1p5`
- `PFHT650_WideJetMJJ950DEtaJJ1p5`
- `PFHT800`
- `PFHT900`
- `PFJet450`

We use a logical "OR" of the following triggers for 2017 and 2018:

- `HLT_AK8PFHT800_TrimMass50`
- `HLT_AK8PFJet330_BTagCSV_p17` (2017F only)
- `HLT_AK8PFJet330_TrimMass30_PFAK8BoostedDoubleB_np2` (2018-only)
- `HLT_AK8PFJet400_TrimMass30` (2017C onward)
- `HLT_AK8PFJet420_TrimMass30` (2017C onward, redundant with above)
- `HLT_AK8PFJet500`
- `HLT_PFHT1050`
- `HLT_PFJet500`

A brief description of the triggers used in the analysis is shown in Table 1.

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Table 1: A description of the triggers used in this analysis and their prescales.

Trigger Name	Definition	Prescale
`HLT_PFHT1050`	AK4PFJets with $p_{\mathrm{T}} > 30\,\mathrm{GeV}$ and $\lvert\eta\rvert < 2.5$ $HT > 1050\,\mathrm{GeV}$	1
`HLT_AK8PFJet[400/420]_` `_TrimMass30`	AK8PFJets with one jet $p_{\mathrm{T}} > 400/420\,\mathrm{GeV}$ and $\lvert\eta\rvert < 2.5$ one jet with trimmed mass $> 30\,\mathrm{GeV}$ trimming parameters: $r = 0.1$ and $pt_{frac} = 0.3$	1
`HLT_AK8PFHT800_` `_TrimMass50`	AK8PFJets with $p_{\mathrm{T}} > 200\,\mathrm{GeV}$ and $\lvert\eta\rvert < 2.5$ and trimmed mass $> 50\,\mathrm{GeV}$ trimming parameters: $r = 0.1$ and $pt_{frac} = 0.3$ $HT > 800\,\mathrm{GeV}$	1
`HLT_PFJet50`	AK4PFJets with one jet $p_{\mathrm{T}} > 500\,\mathrm{GeV}$ and $\lvert\eta\rvert < 5.0$	1
`HLT_AK8PFJet500`	AK8PFJets with one jet $p_{\mathrm{T}} > 500\,\mathrm{GeV}$ and $\lvert\eta\rvert < 5.0$	1
`HLT_AK8PFJet330` `_BTagCSV_p17`	AK8PFJets with one jet $p_{\mathrm{T}} > 330\,\mathrm{GeV}$ and $\lvert\eta\rvert < 2.5$ one jet with trimmed mass $> 30\,\mathrm{GeV}$ trimming parameters: $r = 0.1$ and $pt_{frac} = 0.3$ one jet with CSV> 0.3	1

`HLT_AK8PFJet330`
`_TrimMass30`
`_PFAK8BoostedDoubleB_np2`
	AK8PFJets
	with one jet $p_{\mathrm{T}} > 330\,\mathrm{GeV}$ and $\lvert\eta\rvert < 2.5$
	one jet with trimmed mass $> 30\,\mathrm{GeV}$
	trimming parameters: $r = 0.1$ and $pt_{frac} = 0.3$
	one jet with double-b > -0.2
		1

`HLT_Mu50_v*`	Muon with $p_{\mathrm{T}} > 50\,\mathrm{GeV}$ and $\lvert\eta\rvert < 2.5$	1

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2.2.1 Trigger Efficiency

The efficiency of the trigger soup for each data taking period is measured using a muon reference trigger method. Samples of data collected using the `SingleMuon` primary dataset are analyzed after requiring events to pass a mixture of muon POG-recommended [5] triggers:

- `IsoMu24 OR IsoTkMu24 OR Mu50 OR TkMu50` for 2016 data,
- `IsoMu27 OR Mu50 OR OldMu100 OR TkMu100` for 2017 data, and
- `IsoMu24 OR Mu50 OR OldMu100 OR TkMu100` for 2018 data.

The data events are further required to contain an offline reconstructed muon with $p_{\mathrm{T}} > 25\,\mathrm{GeV}$, $|\eta| < 2.4$, passing the POG loose identification critera, and with PF relative isolation less than 25% of the muon p_{T}. These selections match the muon veto selection used later in this analysis, further details can be found in Section 3.3. This set of events serves as the denominator of the jet trigger soup efficiency measurement in data. For the efficiency measurement in simulation, a collection of HT-binned QCD samples (as enumerated later in this section) is used, with the same jet and muon requirements as in data. As most triggers depend on the kinematics of the leading jet, we parameterize the efficiency as a function of the leading AK8 PUPPI jet, after cleaning jets from any nearby ($\Delta R < 0.8$) muons passing the above requirements. We assume that effects uncorrelated with the leading jet are reasonably modeled by the trigger simulation such that correcting the simulation efficiency as a function of the leading jet will capture the dominant data-to-simulation discrepancies. As some triggers attempt to use b-tagging variables, an explicit check of the efficiency as a function of the DeepDoubleB tagger score was performed, and is shown in Fig. 9, where no significant tagger dependence is observed. This is as expected, since the tagger is a much more powerful discriminator of double-b jets than the HLT algorithm.

The efficiencies of individual trigger soup components per era are shown in Tables 2, 3, and 4 for 2016-8, respectively. Eras where a trigger is unavailable have 0% efficiency, and the luminosity-weighted average is then used to determine the scale factor. Figures 3, 6, and 10 show the efficiency as a function of leading jet p_{T}, and Figures 4, 7, and 11 show the efficiency as a function of leading jet m_{SD} for 2016-8 data and its corresponding simulation, respectively. The resulting 2D efficiency scale factor maps and their statistical uncertainty are shown in Figures 5, 8, and 12, for 2016-8, respectively.

Table 2: Trigger efficiency in data and simulation for the 2016 running conditions. Trigger efficiencies are measured inclusively in AK8 jet kinematic region: $450 \leq p_{\mathrm{T}} < 1000\,\mathrm{GeV}$ and $40 \leq m_{SD} < 300\,\mathrm{GeV}$. All entries have uncertainty less than 1%.

era trigger	MC	2016B	2016C	2016D	2016E	2016F	2016G	2016H
AK8DiPFJet280_200_TrimMass30_BTagCSV_p20	0.93	0.82	0.84	0.86	0.86	0.87	0.89	0.86
AK8PFHT700_TrimR0p1PT0p03Mass50	0.90	0.79	0.79	0.79	0.79	0.79	0.80	0.72
AK8PFJet360_TrimMass30	0.95	0.82	0.82	0.81	0.79	0.79	0.82	0.84
PFHT650_WideJetMJJ900DEtaJJ1p5	0.67	0.50	0.50	0.50	0.49	0.49	0.50	0.45
PFHT650_WideJetMJJ950DEtaJJ1p5	0.64	0.44	0.44	0.44	0.43	0.42	0.44	0.39
PFHT800	0.99	0.91	0.91	0.90	0.89	0.90	0.91	0.00
PFHT900	0.95	0.75	0.75	0.74	0.73	0.73	0.76	0.69
PFJet450	0.81	0.49	0.50	0.49	0.47	0.47	0.49	0.51

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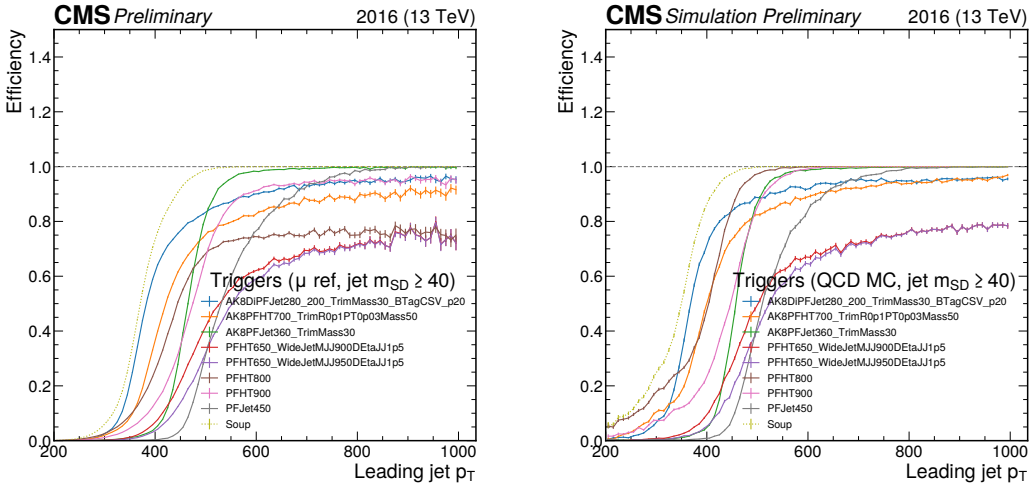

Figure 3: Individual trigger efficiency in data (left) and simulation (right) for 2016 running conditions as a function of AK8 jet p_T.

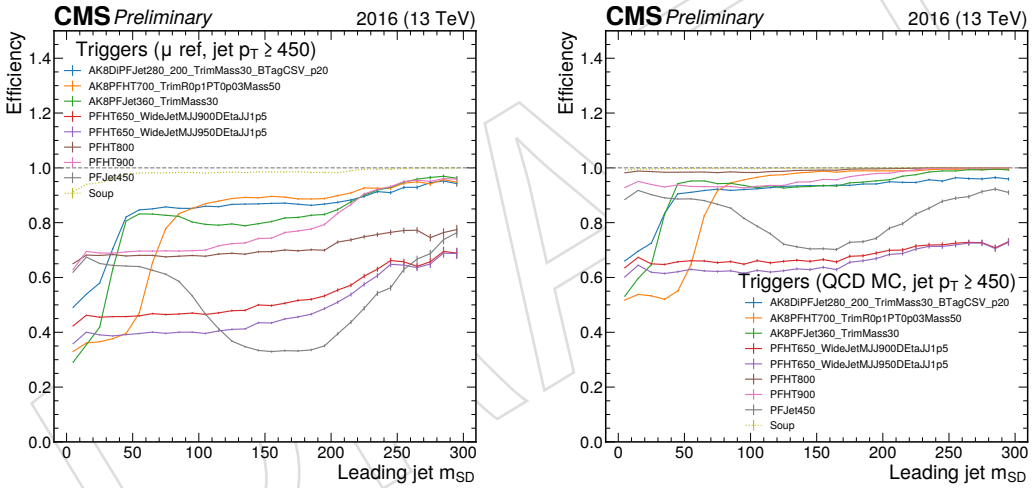

Figure 4: Individual trigger efficiency in data (left) and simulation (right) for 2016 running conditions as a function of AK8 jet m_{SD}.

Table 3: Trigger efficiency in data and simulation for the 2017 running conditions. Trigger efficiencies are measured inclusively in AK8 jet kinematic region: $450 \leq p_T < 1000$ GeV and $40 \leq m_{SD} < 300$ GeV. All entries have uncertainty less than 1%.

era trigger	2017B	2017C	2017D	2017E	2017F	MC
AK8PFHT800_TrimMass50	0.00	0.63	0.63	0.63	0.60	0.86
AK8PFJet330_PFAK8BTagCSV_p17	0.00	0.00	0.00	0.00	0.54	0.94
AK8PFJet400_TrimMass30	0.00	0.64	0.74	0.96	0.94	0.99
AK8PFJet420_TrimMass30	0.00	0.53	0.65	0.91	0.87	0.98
AK8PFJet500	0.48	0.48	0.47	0.46	0.44	0.83
PFHT1050	0.42	0.44	0.43	0.42	0.39	0.81
PFJet500	0.32	0.35	0.33	0.33	0.31	0.72

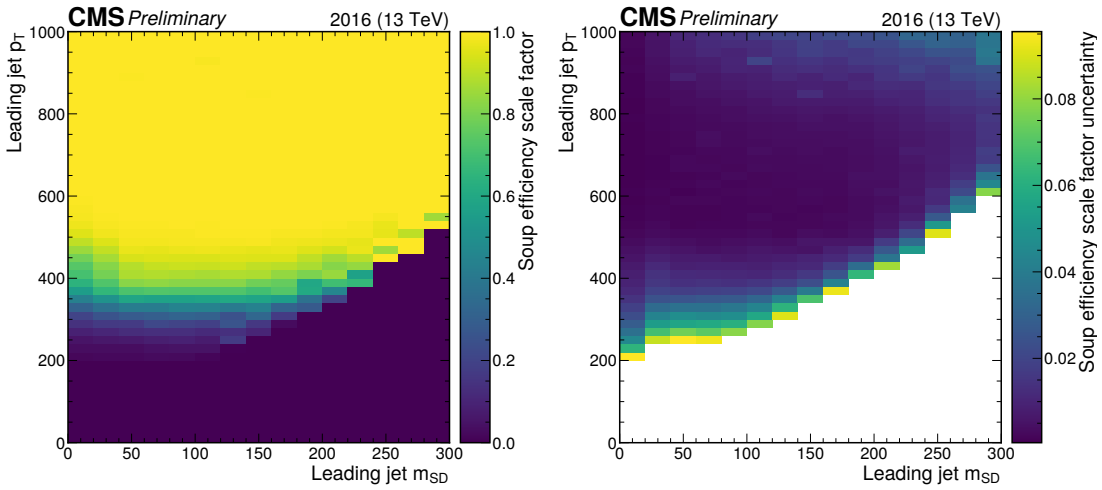

Figure 5: Trigger efficiency scale factor (data to simulation ratio) and its associated statistical uncertainty for 2016 running conditions as a function of AK8 jet (p_T, m_{SD}).

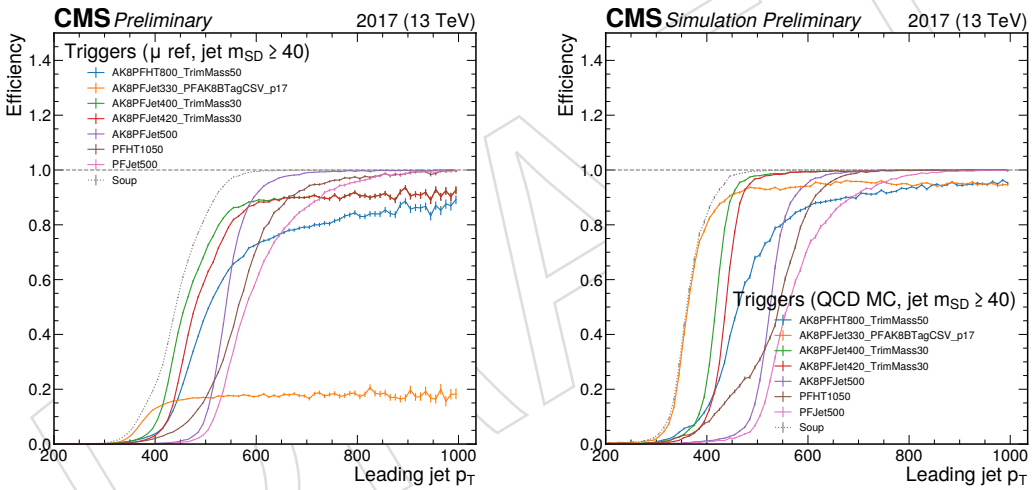

Figure 6: Individual trigger efficiency in data (left) and simulation (right) for 2017 running conditions as a function of AK8 jet p_T.

Table 4: Trigger efficiency in data and simulation for the 2018 running conditions. Trigger efficiencies are measured inclusively in AK8 jet kinematic region: $450 \leq p_T < 1000$ GeV and $40 \leq m_{SD} < 300$ GeV. All entries have uncertainty less than 1%.

era trigger	MC	2018A	2018B	2018C	2018D
AK8PFHT800_TrimMass50	0.84	0.56	0.54	0.54	0.57
AK8PFJet330_TrimMass30_PFAK8BoostedDoubleB_np4	0.79	0.43	0.70	0.70	0.70
AK8PFJet400_TrimMass30	0.99	0.93	0.92	0.92	0.94
AK8PFJet420_TrimMass30	0.98	0.85	0.84	0.83	0.86
AK8PFJet500	0.79	0.41	0.39	0.39	0.42
PFHT1050	0.77	0.38	0.37	0.37	0.39
PFJet500	0.68	0.31	0.29	0.28	0.31

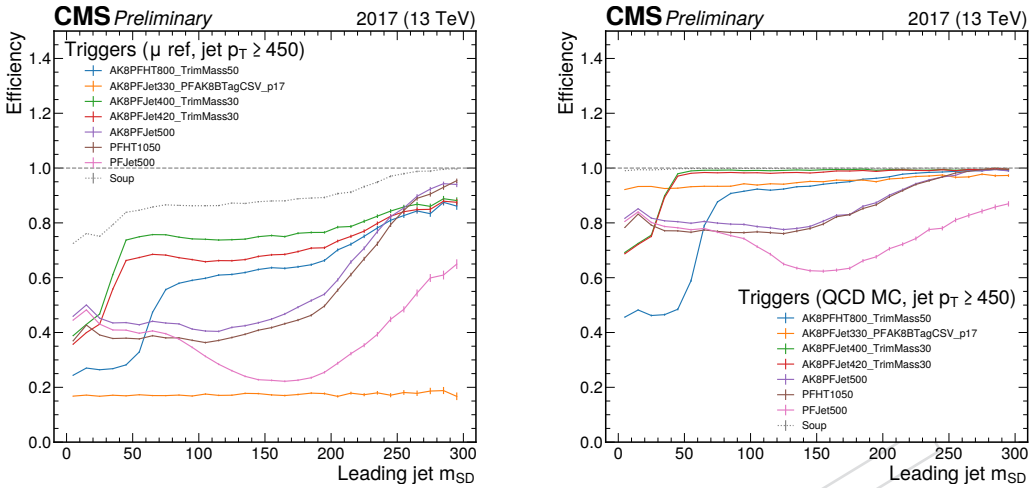

Figure 7: Individual trigger efficiency in data (left) and simulation (right) for 2017 running conditions as a function of AK8 jet m_{SD}.

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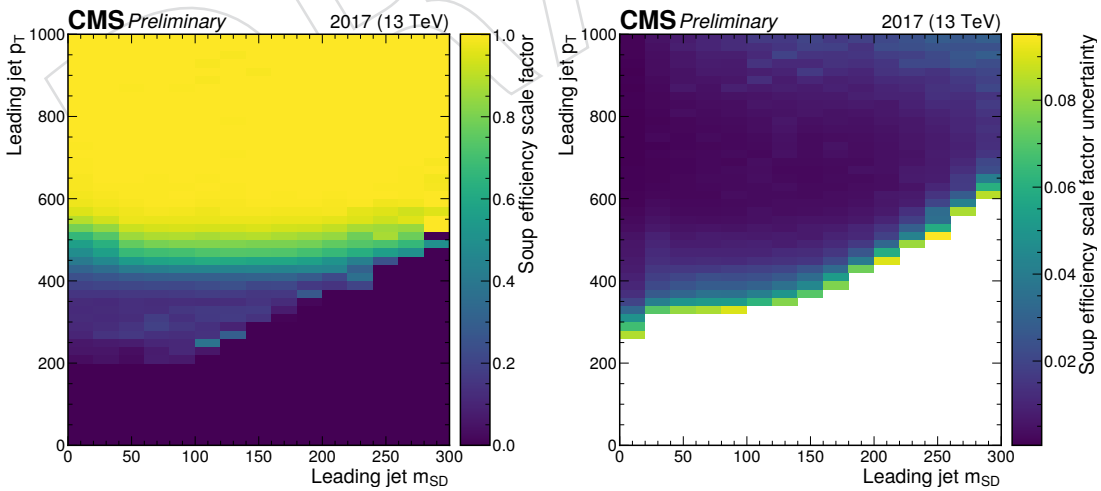

Figure 8: Trigger efficiency scale factor (data to simulation ratio) and its associated statistical uncertainty for 2017 running conditions as a function of AK8 jet (p_T, m_{SD}).

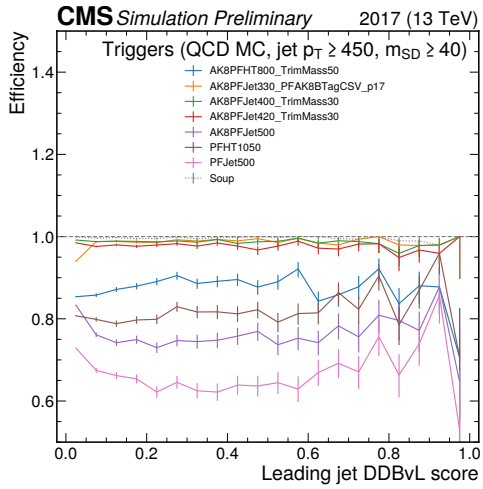

Figure 9: Individual trigger efficiency in simulation for 2017 running conditions as a function of AK8 jet DeepDoubleB tagger score.

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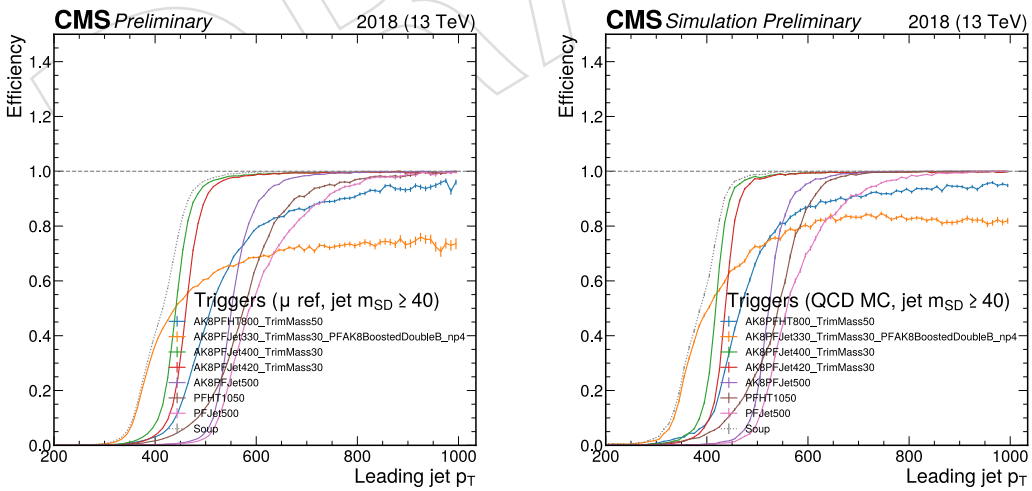

Figure 10: Individual trigger efficiency in data (left) and simulation (right) for 2018 running conditions as a function of AK8 jet p_T.

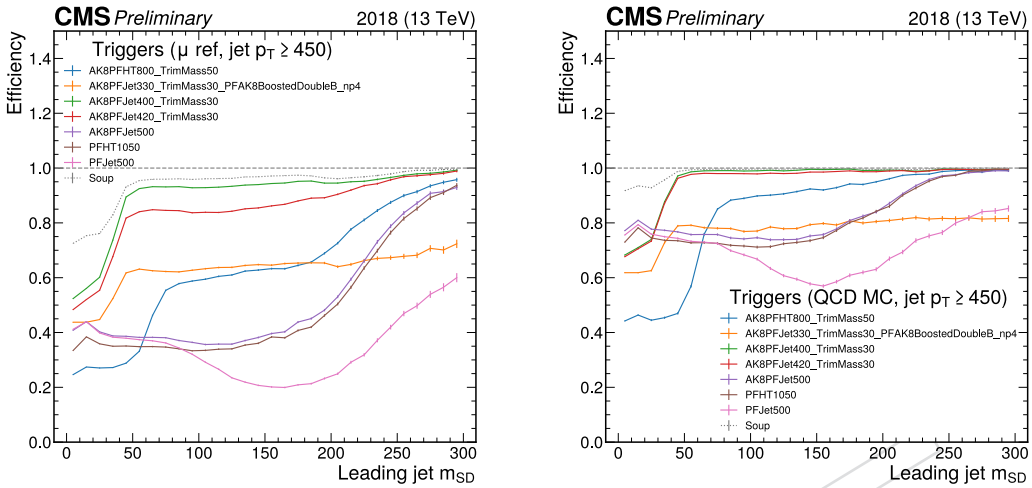

Figure 11: Individual trigger efficiency in data (left) and simulation (right) for 2018 running conditions as a function of AK8 jet m_{SD}.

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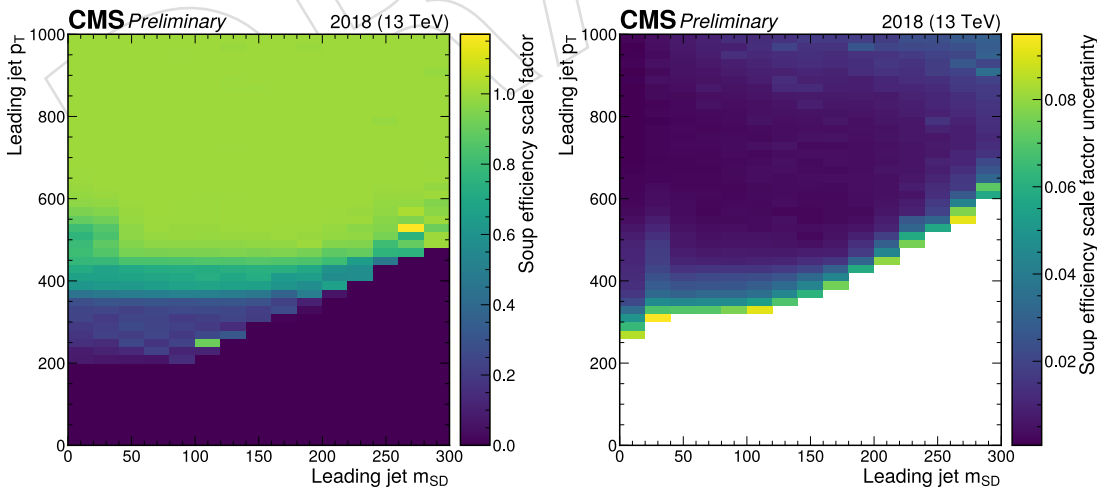

Figure 12: Trigger efficiency scale factor (data to simulation ratio) and its associated statistical uncertainty for 2018 running conditions as a function of AK8 jet (p_T, m_{SD}).